SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ENG201

Timed Online Assignment - January Semester 2023

Linear Systems Analysis and Design

Monday, 15 May 2023 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOACourse Code, Your Name and Student PI No. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT space.
   Example, TOAENG201_RaphaelLee_T1923161 (omit D/O, S/O).

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6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .doc or .docx format. All answers are to be typed in Times New Roman font, size 12 and single spacing. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. Flowcharts and graphs may be scanned or photographed and embedded in the Word file in the right sequence provided it does not exceed the file size limit of 500MB. All uploaded handwritten answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

7. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

8. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

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Answer all questions. (100 marks)

Question 1a

The following signals exist for all time:

- $\cos(3\pi t) + \sin(7\pi t)$
- $\cos(2\pi t) + \sin(3t)$
- $[e^{-5t}]u(t)$
- $[e^{3t}]u(t)$

Question 1a(i)

Analyze which of these signals are periodic and calculate their corresponding periods.

(5 marks)

Question 1a(ii)

Analyze which of these signals are power signals? Assuming they represent voltage signals applied to a 1 Ω resistor, calculate their corresponding average power.

(5 marks)

Question 1a(iii)

Analyze which of these signals are energy signals and calculate their corresponding energies.

(5 marks)

Question 1b

Discuss whether the system described by the following differential equation is linear.

$$\frac{dy(t)}{dt} + y(t) + 4 = x(t)$$

(10 marks)

Question 2a

The impulse response of a linear time-invariant system is given by the following equation:

$$h(t) = \begin{cases} (2-t)u(t), & 0 \le t < 2 \\ 0 \quad , & t \ge 2 \end{cases}$$

If the above system is excited by a unit-step function, describe and sketch the resulting output function in the range $-\infty < t \le 5$.

(10 marks)

Question 2b

The impulse response sequence of a causal, linear time-invariant discrete-time system is given by

$$\{h[n]\} = \{\mathbf{1}, 2, 1, 0, 0, \cdots\}$$

and the input sequence to the above system is given by

$$\{x[n]\} = \{\mathbf{1}, 3, 1, 3, 1, 3, 1, \cdots\},$$

where the boldface indicates the member of the sequences at $n = 0$. Determine the output signal at $n = 0, 1, 2, 3, 4, 5$.

(15 marks)

Question 3a

Calculate the Fourier transform of the sinusoidal sequence

$$x[n] = \cos\{\Omega_0 n\} \quad |\Omega_0| \leq \pi.$$

(7 marks)

Question 3b

A causal discrete linear time-invariant system is described by the following equation

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n],$$

where $x[n]$ and $y[n]$ are the input and output signals of the system, respectively.

Question 3b(i)

Design the frequency response $H(\Omega)$ corresponding to the above given system.

(8 marks)

Question 3b(ii)

Solve and obtain the impulse response $h[n]$ of the above given system.

(10 marks)

Question 4a

A continuous linear time-invariant system is described by the transfer function

$$H(s) = \frac{s+3}{s^2+3s+2}.$$

The above system is excited by an input signal:

$$x(t) = [e^{-0.5t} + 3e^{-1.5t} - 5\,]u(t),$$

where $u(t)$ is the unit-step function. Determine the steady state response of this system using the Laplace transform. Assume all initial conditions to be zero.

(15 marks)

Question 4b

The transfer function of a linear discrete-time system is given by the following expression:

$$H(z) = \frac{z^2(z+0.5))}{0.2z^3 + 0.4z^2 + 1.15z + 0.5}$$

The input sequence to the above system is given by $\{\mathbf{0.2}, 0.3, 1, 0, 0, \ldots\}$, where boldface indicates the member of the sequence at $n = 0$.

Question 4b(i)

From the above description of the system, find the output sequence.

(6 marks)

Question 4b(ii)

As a consequence of your result from Question 4b(i), can you simplify the given transfer function, $H(z)$?

(4 marks)

----- END OF PAPER -----